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Review Article

User-Centered Evaluation of Explainable Artificial Intelligence (XAI): A Systematic Literature Review

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Researchers have developed a variety of approaches to evaluate explainable artificial intelligence (XAI) systems using human–computer interaction (HCI) user-centered techniques. This systematic literature review has been conducted to understand how these approaches are used to achieve XAI goals. The aim of this review is to explore the methods used to evaluate XAI systems in studies involving human subjects. A total of 101 full-text studies were systematically selected and analyzed from a sample of 3414 studies obtained from four renowned databases between 2018 and 2023. The analysis focuses on prominent XAI goals achieved across 10 domains and the machine learning (ML) models utilized to create these XAI systems. The analysis also explores explanation methods and detailed study methodologies used by researchers in previous work. The analysis is concluded by categorizing the challenges experienced by researchers into three types. Exploring the methodologies employed by researchers, the review discusses the benefits and shortcomings of the data collection methods and participant recruitment. In conclusion, this review offers a framework that consists of six pillars that researchers can follow for evaluating user-centered studies in the field of XAI.

1. Introduction

The use of artificial intelligence (AI) in everyday life has noticeably increased throughout the past decade, in areas such as medicine [1], education [2], and eye tracking [3]. During the COVID-19 pandemic, unexplained AI system prediction trends of the disease were shown to the public [4]. These predictions required exploiting comprehensive data, such as hospital settings, capacity, the number of COVID-19 tests taken daily, demographics, population density, and the vulnerability of each individual in that population. But patterns of a new disease can never be expected as the actual pattern proved the inaccuracy of these predictions [4]. Thus, user trust in AI systems became a massive challenge for machine learning (ML) experts. For human beings to better trust AI systems' judgments, users must be presented with explanations that enhance their understanding of the AI systems [5]. Considering this, explainable artificial intelligence (XAI) should not be viewed as only a need but as a requirement. Mohseni et al. [6] stated that improving users' understanding and enhancing users' trust is the ultimate approach to making XAI systems more acceptable and trustworthy.

The need for system explanations surfaced with the first wave of AI implementation [6]. In 1977, authors stressed the need for system explanations, especially for nonexperts in the industry [6]. System explanations increase AI system's credibility while allowing users to learn from the explanation [6]. The idea of system explanations was revived in 2017 with the introduction of the XAI program created by the Defense Advanced Research Project Agency (DARPA) [7]. The XAI program led to XAI becoming integral to various domains, including medicine, transportation, law, and finance [7]. XAI seeks to develop ML techniques that produce explainable models while not interfering with the learning performance of AI techniques used to evaluate data [7]. Through the use of XAI, researchers anticipate increasing human trust in ML, allowing end users to understand the outcomes of complex AI systems, which are known as black boxes [8].

There is a high potential for XAI in various fields [9]. Given that each industry has a unique set of target audience, it is critical to comprehend the efficacy of XAI solutions and how end users perceive them [10]. This is where human–computer interaction (HCI) comes into play [10]. According to Nazar et al., the gap in user-centered research between AI and HCI has begun to close as research trends in human AI (HAI), usable AI (UAI), and XAI [10]. Understanding XAI user requirements, on the other hand, has still to be investigated and improved [10]. While two recent reviews have analyzed how XAI and HCI are collaboratively implemented in an interdisciplinary manner, one review focuses on one domain and the other on a few domains [9, 10]. These reviews are missing studies that explore user-centered approaches to evaluate explainability in all domains in which XAI has been implemented.

This review examines user-centered evaluation approaches in HCI research for XAI systems by analyzing 101 full-length research papers published in various venues from 2018 to 2023. Systematic reviews in the field have integrated existing knowledge on different aspects of XAI, including the progress made in XAI and knowledge domains [11] and the potential of XAI in healthcare [12]. Understanding how XAI is evaluated through user-centered approaches is crucial in acknowledging the significance of end users, who are central to the development of such technology. The goal is to accumulate knowledge from a substantial collection of studies, focusing on prominent XAI goals and domains explored by researchers. The review also explores the utilization of ML models and explanation methods to achieve XAI goals, while investigating the contribution of HCI evaluation techniques and classifying challenges faced by researchers. Furthermore, the review provides recommendations for future researchers conducting user-centered evaluations, enabling critical assessment of different evaluation approaches, and supporting researchers in their evaluation of future studies.

This work begins with positioning the review within the broader context of background and related work in the field of XAI, considering previous reviews carried out by researchers in this area (Section 2). Then, the methods used for this systematic literature review are presented (Section 3). Next, a description of evaluation approaches found in different domains to achieve XAI goals is provided (Section 4). In the Discussion section, an analysis of the choice of data collection methods and how it is driven by the domain and the XAI goal, participants recruited in studies, and the selection of XAI and representation methods and how these factors are correlated to user expertise is explained (Section 5) before providing the six-pillar framework (Section 6) and concluding (Section 9).

2. Background and Related Work

2.1. XAI. AI has become an integral part of using technology. The General Data Protection Regulation (GDPR) commission in the European Union has established users' rights to explanations of privacy, data protection, and transparency of AI system algorithms. Because of this, every AI system is expected to be explainable [13]. Black boxes are systems that use AI logic for predictions and making decisions. However, as a black box, it is almost impossible for humans to understand how the system reached a decision or a prediction, nor understand the level of data security and safety of such systems [14]. This is a critical matter, as explained by the GDPR commission in the European Union [13]. The reason is that fields such as personalized medicine, robotic assistants, and self-driving cars cannot risk the possibility of making wrong decisions [15]. Guidotti et al. have identified several black box problems such as transparency, outcome explanation problems, and model inspection problems, which allow individuals to understand the logic behind the decision or the prediction [15]. So, XAI comes to open the black box to *comprehend* the internal procedure, explain the outcome, and make it more transparent using terms and visuals that are shown on the user interface (UI).

Various research domains tend to look at the concept of XAI from different perspectives. For example, visual analytics researchers focus on visualizing black boxes to study how users interact with visual XAI. In contrast, psychology researchers are more interested in humans' understanding of the system and how it allows them to develop a better explanation structure. However, researchers in the HCI domain focus on end users and their needs, including enhancing their experience with AI explanations. This subsection illustrates concepts related to XAI that researchers implement in their work.

2.1.1. Explainability of a Model. Explanations can be classified according to their methods of interpretation, which are global and local. A global explanation describes how a ML model works, including the visual designs of the feature contributions [16, 17]. Local explanations focus on explaining how two attributes in a ML model are related. A local explanation also explains individual queries that users have requested, such as saliency maps [18]. Mohseni et al. [6] demonstrated that local explanations suit novice users more than experts. However, experts can use this explanation method to test edge cases when developing a system [6].

2.1.2. XAI Representation. The representation of XAI comes in two forms: visual and textual explanations. A visual explanation uses graphic elements to explain the results that a ML model provides [6], such as the gradient-weighted class activation mapping (Grad-CAM) technique [19] and saliency maps [20], and a dashboard of scores such as confidence and sensitivity of a model's decision [16]. Textual explanations provide reasoning in text, such as words, sentences, or natural language [6]. Many recommendation systems implement this form of representation [21], including some XAI prediction systems.

2.2. The DARPA. It is crucial to highlight the DARPA, a research and development agency based in the United States [7]. DARPA is the key contributor and driving force behind the expansion of research in the field of XAI. As new AI capabilities emerged, DARPA initiated a 4-year XAI program that took effect in 2017. The program focuses on a

system's ability to explain its own decisions, making it convenient for lay users to understand and trust the system and manage stakeholders. This XAI program shed light on the importance of designing and performing user studies to evaluate explanations, enhance user trust, and encourage the appropriate use of AI systems, enhancing the collaboration between humans and machines. The DARPA project has supported researchers in exploring approximately 20 technical approaches to XAI. Fifty articles, all open-access publications, have been published using DARPA project funding, including one in this review [22]. The increasing interest in the field of XAI has led to a massive wave of XAI research exploring enhancing its capabilities and ensuring constant improvement. In addition, many researchers have conducted reviews and surveys on XAI studies.

2.3. Reviews and Surveys on XAI. In the past few years, many reviews have been published covering the research done in the XAI field. For example, a systematic review was conducted in 2019 to explore the need for XAI [23]. Ayodele and Yusoff [23] carried out a critical analysis on four thematic debates in this field, contributing to the knowledge in XAI and future work. A review paper published in the proceedings of the 2018 HCI Conference [24] was aimed at identifying future directions and opportunities for HCI researchers and highlighting the importance of these contributions. Through analysis of 289 papers and 12,412 citing papers, the review mapped the interconnectedness among research domains, ML models, explainable systems, and forming research communities [24]. It found a gap between HCI and the later in the development of explainable systems [24]. Exploring a single domain, Kaadoud et al. [11] reviewed recent progress and interconnection between XAI and knowledge domain. This review concludes that incorporating aspects of knowledge contributes to delivering transparent black boxes [11]. These aspects include knowledge discovery, representation, and representation learning [11]. Hu et al. [25] developed a framework that classifies AI explanations based on six aspects: the purpose of a system, explanation method, context of the explanation, explanation format, stakeholders, and domain. Korica et al. [12] explored XAI opportunities, challenges, and gaps in the healthcare field. This review presents a novel approach in addressing XAI problems within the domain of healthcare [12]. By defining well-identified problems, suitable explanation models can be found that align with the clinical workflow and consider all relevant attributes [12]. Moreover, Chromik and Butz [26] discussed UI design principles for XAI systems, providing a valuable reference for designers to inform their UI development. This review emphasizes the importance of designing user-friendly, human-centric XAI systems [26]. Collectively, these reviews and studies have significantly contributed to the understanding and advancement of XAI.

Looking at surveys, Guidotti et al. [15] classified black box problems into four categories, which are a model explanation, outcome explanation, model inspection, and transparent box design problem. They then developed five XAI system evaluation measures to consider during the design process to avoid the issues categorized in the survey [15]. These evaluation measures include a mental model of how users will understand the system and how it works, explanation usefulness and satisfaction measures, user trust and reliance measures, measuring HAI task performance, and computational measures of the correctness and completeness of the techniques used for explanations [15]. Next, Mohseni et al. [6] conducted a survey on XAI design goals and evaluation methods that could be implemented in multiple areas, such as ML, HCI, and visualization. This survey proposed a user-dependent XAI framework with guidelines for specific design stages, implementing tailored XAI goals based on user expertise [6]. In their work, Ayodele and Yusoff [23] addressed XAI methods in medical decision-making, highlighting challenges and opportunities for future research. In another survey exploring the medical domain, Tjoa and Guan [27] explored the application of interpretability methods in this domain and how clinicians and practitioners can benefit from these methods. This review also sought to understand the appropriateness of these interpretation methods in the medical domain, as well as to encourage the use of such concepts for medical education [27]. Furthermore, Sahakyan et al. [28] conducted the first survey on implementing XAI techniques with tabular data, widely used in fields like criminal justice, healthcare, and finance.

In conclusion, the multitude of published reviews and surveys in the XAI field over the past few years has significantly contributed to advancing the understanding of XAI systems. These works have explored various aspects, including the need for XAI, interconnections among research domains, incorporating knowledge, classification frameworks for AI explanations, UI design principles, and the application of XAI techniques in specific domains. They have also provided valuable insights, future directions, and identified gaps, informing researchers, practitioners, and designers in their pursuit of transparent and accountable AI systems.

2.4. Funded Projects. This review explores several studies that various agencies have funded or that are parts of bigger projects that are grant-funded. The DARPA XAI program has funded many projects. A National Science Foundation (NSF) grant-funded project that Nourani et al. [29] moderated focuses on XAI in relation to video analysis. In addition, the NSF funded three conference proceedings, two extended abstracts, and one workshop. Detailed information about this published work is shown in Table 1.

An ongoing video analysis study focuses on XAI motion capture in recorded stroke rehabilitation exercises [34]. This project receives support from the NSF and the IntelligentCare project, which receives funding from Carnegie Mellon University in Portugal. Five publications of this ongoing work, which ends in 2023, exist, as shown in Table 2.

Another project by Prange et al. [40] focuses on digitizing standardized tests and developing automatic cognitive assessment tools that patients can use to conduct neuropsychological testing on themselves. The cognitive assessment

Table 1: Explainable video analysis project.

Publication title	Publication year	Authors	Publication venue
"Anchoring Bias Affects Mental Model Formation and User Reliance in Explainable AI Systems"	2021	[30]	26th International Conference on Intelligent User Interfaces (IUI '21)
"Investigating the importance of first impressions and explainable ai with interactive video analysis"	2020	[29]	Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems
"The role of domain expertise in user trust and the impact of first impressions with intelligent systems"	2020	[31]	Proceedings of the AAAI Conference on Human Computation and Crowdsourcing
"Soliciting human-in- the-loop user feedback for interactive machine learning reduces user trust and impressions of model accuracy"	2020	[32]	Proceedings of the AAAI Conference on Human Computation and Crowdsourcing
"Explainable Activity Recognition in Videos"	2019	[33]	In Intelligent User Interfaces (IUI) Workshops

Table 2: Explainable AI and stroke rehabilitation project.

Publication title	Publication year	Authors	Publication venue
"A Human-AI Collaborative Approach for Clinical Decision Making on Rehabilitation Assessment"	2021	[35]	Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems
"Co-Design and Evaluation of an Intelligent Decision Support System for Stroke Rehabilitation Assessment"	2020	[36]	Proceedings of the ACM on Human-Computer Interaction
"An exploratory study on techniques for quantitative assessment of stroke rehabilitation exercises"	2020	[37]	Proceedings of the 28th ACM Conference on User Modeling, Adaptation and Personalization
"Interactive hybrid approach to combine machine and human intelligence for personalized rehabilitation assessment"	2020	[38]	Proceedings of the ACM Conference on Health, Inference, and Learning
"Learning to assess the quality of stroke rehabilitation exercises"	2019	[39]	Proceedings of the 24th International Conference on Intelligent User Interfaces

tool analyzes user performance and explains the test results before visiting a doctor. This work received funding from the Federal Ministry of Education and Research in Germany. This research is also part of a project called Interakt and two other projects that the Oldenburg University in Germany managed. Publications about this project are listed in Table 3.

3. Research Approach

To develop a well-designed XAI system that aligns with the needs of end users, it is crucial for designers to identify a clear goal and use the appropriate measures to evaluate it. This systematic literature review is aimed at categorizing studies in the field of XAI based on the XAI goals identified by Barredo Arrieta et al. [8]. These XAI goals are established by designers in accordance with the expectations of the target audience from the system, thus contributing to the fulfilment of one of the capabilities of XAI [8]. However, only a limited number of studies adopt a conceptual perspective in defining XAI goals [8]. Thus, Barredo Arrieta et al. [8] have defined the following XAI goals as classification criteria for the development of XAI models in the field of XAI.

In the field of XAI, researchers strive to achieve at least one of the nine significant XAI goals [8]. Trustworthiness is a key consideration, as individuals expect the system's decisions to align with certain desired behaviors. Causality is aimed at establishing the relationships and causal effects between variables in the XAI system. Transferability explores the potential for reusing knowledge from other domains to enhance system explanations, while informativeness focuses on providing users with sufficient information that they can relate to in decision-making, prediction, and recommendation systems. Confidence relates to the reliability and stability of the system, directly impacting its trustworthiness. Fairness emphasizes the importance of visualizing variable relationships to address biases in ML models and prevent unethical use of model outputs. Accessibility is aimed at making the information generated by internal algorithms understandable, particularly for nontechnical users. Interactivity is crucial in systems where user input significantly influences system behavior, allowing users to interact with and adjust the model. Lastly, privacy awareness highlights the need to evaluate the privacy implications and potential breaches associated with using sensitive data in XAI explanations. By addressing these goals, XAI researchers aim to create systems that are trustworthy, interpretable, fair, accessible, and privacy-conscious.

3.1. Research Objectives. This review is aimed at extracting knowledge from the XAI research conducted between 2018 and 2023. The review provides an overview of XAI goals

Table 3: Explainable AI and automatic cognitive assessment project.

Publication title	Publication year	Authors	Publication venue
"Modeling Cognitive Status through Automatic Scoring of a Digital Version of the Clock Drawing Test"	2019	[41]	Proceedings of the 27th ACM Conference on User Modeling, Adaptation and Personalization (Larnaca, Cyprus) (UMAP '19)
"Multimodal Speech-based Dialogue for the Mini-Mental State Examination"	2019	[24]	Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems, CHI 2019
"A categorisation and implementation of digital pen features for behaviour characterization"	2018	[42]	Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)
"Learning classification models of cognitive conditions from subtle behaviors in the digital Clock Drawing Test"	2016	[43]	Machine Learning
"The clinical data intelligence project"	2016	[44]	Informatik-Spektrum
"Intelligent cognitive enhancement technology by cognitive models and mixed reality for dementia patients"	2015	[45]	AAAI Fall Symposium Series
"Integrating digital pens in breast imaging for instant knowledge acquisition"	2014	[46]	AI Magazine

and explores how they are implemented and evaluated from an HCI perspective. The primary goals of this review are as follows:

- To assess the role of HCI evaluation techniques in achieving XAI goals.
- To examine the evaluation methods employed for XAI goals in HCI.
- To explore the implementation of XAI goals.
- To ascertain the emphasis placed on specific goals in research.
- To examine the domains where XAI goals are applied.
- To review the ML models utilized for various XAI goals.
- To identify the challenges in implementing XAI goals across different fields.

Thus, the review is aimed at answering the following questions:

- RQ1: What goals are researchers implementing to achieve a well-presented XAI?
- RQ2: Which XAI goals do researchers focus on the most?
- RQ3: In which domains are researchers aiming to implement XAI goals?
- RQ4: Which ML models are used to deliver different XAI goals?
- RQ5: How is incorporating HCI techniques contributing to presenting better XAI goals?
- RQ6: How are XAI goals evaluated in the field of HCI?
- RQ7: What are the challenges of implementing XAI goals in various domains from an HCI perspective?

RQ1–3 cover the range of XAI goals that researchers in many domains address and aim to deliver a well-presented XAI. RQ4–7 focus on XAI in relation to HCI, exploring the potential of the interconnection between both fields.

3.2. Review Process. The selection of studies for the systematic literature review went through four phases, beginning with identifying studies. The second phase involved filtering studies in two steps: (1) reading the title and abstract and excluding irrelevant articles and (2) reading the full texts and excluding studies for different reasons discussed in later paragraphs. The third phase was to extract data from the selected studies after the full texts were read. Lastly, data extraction was done to include the texts in the fourth phase, providing the results, where all the data was conveyed and discussed. The PRISMA flow diagram shown in Figure 1 summarizes the selection process up until the data extraction of the final studies included in this review. Also, a supporting *PRISMA checklist* reports standards adhered to in the systematic review conducted for this study to ensure completeness in reporting the review process and findings.

3.2.1. Phase 1: Identification of Relevant Studies. In accordance with PRISMA guidelines, the following steps have taken place to identify relevant studies for this systematic literature review: First, a search query was explored in four different databases. Obtaining studies from different databases resulted in having duplicate studies that were eliminated before the screening phase. These studies were also evaluated against a certain criterion, which is described in the following section. The following steps have been followed to identify relevant studies for this review. First, a search query was explored in four different databases. Then, duplicate studies

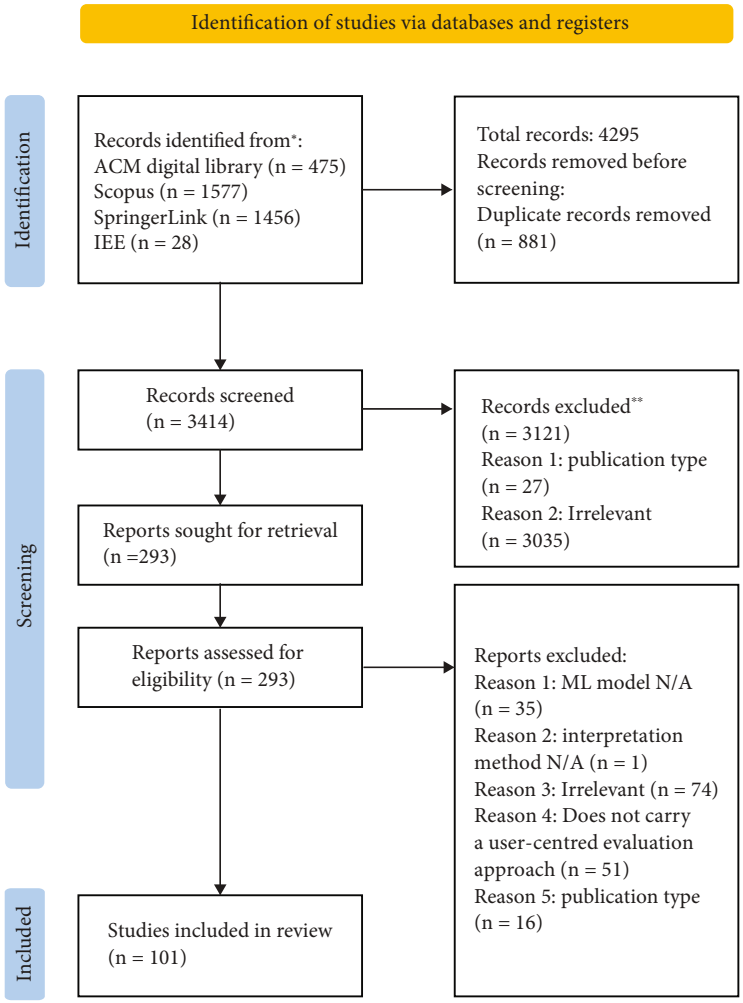

Figure 1: Procedural flow chart following PRISMA guidelines.

were eliminated before the screening phase. The studies were evaluated against a criterion that is described in this section.

3.2.1.1. Search Procedure. The literature search was completed on January 21, 2023. An advanced search of keywords was used to obtain systematic results of potentially relevant work between 2018 and 2023 from the Association of Computing Machinery (ACM) Digital Library, Scopus, SpringerLink, and the Institute of Electrical and Electronics Engineers (IEEE). The interest of this review was to explore studies with a clear focus on XAI and XAI goals achieved in a user-centered approach that interconnects with HCI. The following search query was used: *(XAI OR "explainable AI" OR explainabl ∗ OR "explainable artificial intelligence") AND ("human computer interaction" OR HCI OR "human artificial intelligence" OR "human AI" OR "human-ai interaction" OR "user experience" OR UX) AND ("machine learning" OR "deep learning").* Queries were constructed differently for each database to ensure the broadest range of results. The search query resulted in 4295 papers and was reduced to 3414 after removing duplicates. Table 4 shows the detailed search query results from each database explored in this review. In addition, the selected articles had to meet the criteria described in Table 5. Ultimately, 101 publications were selected for this systematic literature review.

Table 4: Databases explored in this review.

Database/publication type	No. of papers
ACM Digital Library	
Proceedings	583
Journal articles	200
Scopus	
Conference papers	818
Articles	750
Book chapters	35
SpringerLink	
Conference papers	488
Book chapters	897
Articles	482
IEEE Xplore	
Excluding magazines	43

Table 5: Eligibility criteria.

Inclusion	Exclusion
Empirical studies	Duplicate studies
Studies published in peer-reviewed journals	Review articles
Proceedings	Papers that are irrelevant to XAI
Book chapters	Studies in any language other than English
Studies published in English	Books
Studies published from 2018 to 2023	Workshop papers, courses, tutorials, and talks

3.2.2. Phase 2: Filtering Studies. In Phase 2, the studies retrieved from the four databases were filtered twice to identify the papers relevant for this review. The first step was to exclude studies based on general criteria of exclusion that could be identified from the title and abstract of each study. The second step of study filtering is reading the full texts to evaluate studies against more specific exclusion criteria. By the end of these two phases, 101 studies had been selected and analyzed in this review.

3.2.2.1. Step 1: Filtering Initial Set of Studies. The title and abstract of 3414 publications were reviewed, and studies that met the following exclusion criteria were excluded. All papers that met one of the exclusion criteria were coded. Some exclusion criteria were difficult to identify in several publications by reading only the title and the abstract. These papers were transferred, along with the papers that were included, into Step 2, where the author read the full texts. The first step of review resulted in removing 3121 publications and allowing 293 to advance to *Step 2* of Phase 2. The exclusion criteria are as follows:

- Studies that were review articles, workshop papers, tutorials, books, newspapers, magazines, proposals, extended abstracts, posters, Ph.D. dissertations, or MSc theses.
- Studies that were irrelevant to the scope of this review.
- Studies that did not use a user-centered approach as an evaluation method.
- Studies in any language other than English.

3.2.2.2. Step 2: Another Stage of Filtering Studies. At this stage, 293 full texts were reviewed, and 188 studies were excluded for the following reasons:

- Studies where a ML model was used, but not clearly identified in the papers ($n = 35$). These studies were excluded because this study aims to explore whether a pattern exists where specific XAI goals are achieved via specific AI models
- Studies where the explanation method could not be identified ($n = 1$)
- Studies with data about the evaluation methodology that is missing ($n = 1$)
- Studies that were review articles, workshop papers, tutorials, books, newspapers, magazines, proposals, extended abstracts, posters, Ph.D. dissertations, or MSc theses ($n = 16$)
- Studies that were irrelevant to the scope of this review ($n = 70$)
- Studies that did not use a user-centered approach as an evaluation method ($n = 51$)
- Studies in any language other than English ($n = 8$)
- Studies that were duplicates ($n = 6$)

3.2.3. Phase 3: Detailed Review of Studies. The author reviewed the remaining 101 papers after screening all studies retrieved from 4 databases based on the search query in *Section 3.2.1.1*, which are shown in Table 4, extracting data concerning the goal of this study's analysis from each document and entering it into a spreadsheet. The analysis includes different attributes that contribute to answering the research questions mentioned above. These attributes are as follows:

- Publication venue (*Section4.1.1*)
- Country of publication (*Section4.1.2*)
- Publication by year and country (*Section4.1.3*)
- Publication type (*Section 4.1.4*)
- XAI goals addressed in the reviewed domains (*Section 4.1.5*)
- ML models used to achieve XAI goals (*Section 4.1.6*). This was classified based on the survey that lists the black boxes that were opened in the papers they reviewed [47]
- Explanation methods (*Section 4.2*). This was classified as local and global explanations [48]
- Detailed procedure of included studies (*Section 4.3*)
  - o. Research methods (*Section 4.3.1*)
  - o. Study design (*Section 4.3.2*)
  - o. Participants: number of participants, their expertise (*Section 4.3.3*)
  - o. Data collection method (such as questionnaires and interviews, *Section 4.3.4*). Evaluation tools used (*Section4.3.5*)
  - o. Medium of evaluation, which classifies the different interfaces used to evaluate XAI systems (*Section 4.3.6*)
  - o. Analysis methods (*Section 4.3.7*)

- Challenges (*Section 4.4*). Challenges were classified in this review based on observing challenges that are scope-related, methodology-related, and user-related

3.2.4. Phase 4: Discussion. This work explored the research trends and observations conveyed in the papers reviewed. This analysis specifically focused on XAI research trends, the goals of the trends, and how HCI was incorporated to achieve the various XAI goals. The results of this analysis are presented in Section 4.

3.3. Challenges Encountered in This Review. This review is based on analyzing 101 papers that were selected after screening 3414 studies, a large corpus of papers. N.A. independently reviewed and coded the papers and then discussed them with a D.A. who is a domain expert to resolve any disagreement or confusion regarding including or excluding a study. In addition, Cohen's kappa inter-rater agreement was run on by D.A. and R.A., who are domain experts, to avoid biases in selecting studies [49]. Running Cohen's kappa test requires reaching an agreement that achieves a result of at least $k = 0.8$, to ensure that N.A. is not biased in the selection of studies to be included [49]. The two experts reviewed 20% of the total studies to be reviewed in Phase 1 (683 out of 3414). The experts followed the selection criteria for the inclusion and exclusion of studies. Based on the test results, the three researchers reached an agreement on 80.8% of the 683 articles, close to Cohen's kappa least agreement.

Furthermore, despite the fact that the need for explanation in AI systems was noted as early as 1977 [6], the decision to focus primarily on recent studies between January 2018 and January 2023 was made. The rationale for this is that DARPA's XAI program began in 2017, and journal publications of these studies began in 2018 [7]. As a result, there has been a significant rise in study in this area over the last 6 years.

4. Results

A mixed methods synthesis, which combines quantitative and qualitative analysis, was performed on the data extracted from the included studies [50]. First, the analysis provides an overview of the datasets and XAI goals researchers aimed to achieve in the selected studies (RQ1). Then, the XAI goals researchers focused on the most were explored (RQ2). The analysis also describes the domains that researchers concentrated on and the objectives to achieve (RQ3). In addition, a description of the user-centered approach and HCI techniques used to evaluate the XAI goals is provided (RQ4). Next, the evaluation methods used in each study are conveyed (RQ5). By the end of this section, light is shed on the challenges of implementing XAI goals in various domains from an HCI perspective (RQ6).

4.1. Data Description. This section presents an overview of the dataset, providing descriptive statistics regarding the included studies, such as their distribution across various years and publishing venues. Additionally, it offers high-level information about the selected studies, including their country of publication and type of publication. Moreover, XAI has been implemented in different domains to address XAI goals that are explored in this section. The section also identifies the goal of each study and explains how the goal was achieved through the study's objective. Next, it explores the ML models used to achieve different XAI goals, aiming to identify any associations between XAI goals and the ML models used. After that, a detailed description of the user-centered approaches to evaluating the XAI systems from an HCI perspective and a discussion of XAI explanation methods is provided. This section is concluded by discussing the challenges that each study encountered, which have been classified into three categories.

In short, the dataset included a total of 101 studies, with a majority published in 2021 ($n = 45$), followed by 2022 ($n = 26$), 2020 ($n = 13$), 2019 ($n = 10$), 2023 ($n = 7$), and 2018 ($n = 1$). A significant number of studies related to the topic of the review were conducted in the United States and published in conference papers, especially in ACM journals and conferences such as the International Conference on Intelligent User Interfaces (IUIs), the Transactions on Interactive Intelligent Systems journal, and the CHI Conference on Human Factors in Computing Systems. The authors of the studies aimed to achieve various XAI goals, with the most frequent goal being trustworthiness. In terms of ML models used to develop XAI systems, deep learning (DL) models and classification models were extensively used in the selected studies to achieve different XAI goals. The studies used various XAI representation methods to explain the system, which are explored in this review. This work also investigates the user-centered approaches followed by researchers to evaluate XAI systems and examine the achievement of different XAI goals. This section concludes with a discussion of the challenges encountered in the selected studies.

The articles included in this study target different XAI domains. Out of 101 studies, 32 were published to evaluate XAI without specifying a specific domain (29.7%), while 15 studies focused on the medical domain (14.8%). In addition, 13 studies focused on economics (12.8%), and 11 studies discussed the education domain (10.8%). Moreover, 6 studies explored the social media domain (5.9%), 19 studies were conducted in the entertainment domain (18%), 2 studies looked at the recruitment domain (1.9%), and 6 studies investigated the legal domain (7.9%). The total distribution of the articles analyzed in this study is shown in Figure 2. The publication frequency across the years is shown in Supporting Information Appendix A.

4.1.1. Publication Venues. The studies included in this review were published in 60 different publication venues: 15 in the Conference Proceedings of IUIs; 10 in ACM Transactions in the *Interactive Intelligent Systems* journal; 10 in Proceedings of the CHI Conference on Human Factors in Computing Systems; 5 in the *Journal of Artificial Intelligence (AIJ)*; 4 in the ACM Conference on User Modeling, Adaptation and Personalization (UMAP); 3 in Central Europe (CEUR) Workshop Proceedings; 3,

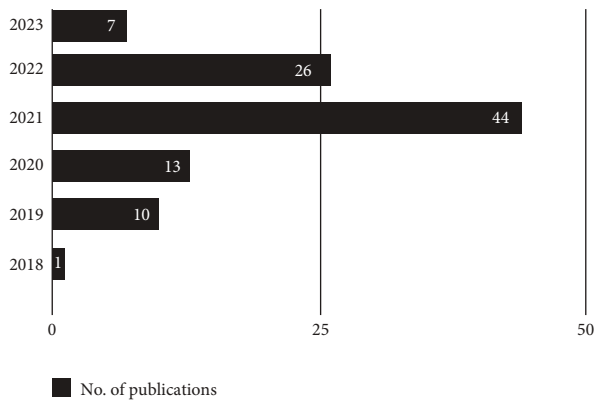

Figure 2: Number of publications of included studies in different years.

which were initially conference papers, in the book sections of *Lecture Notes in Computer Science*; and 3 in the Proceedings of ACM HCI. Regarding the other 40 studies, 1 to 2 studies were published in different publication venues (see Supporting Information Appendix B).

4.1.2. Countries of Included Studies. As shown in Figure 3, the studies included in this review were conducted in 19 countries. The countries that performed user-centered studies in the field of XAI most frequently during the past 5 years were the Unites States (39.6%), Germany (22.7%), Canada (5%), and Singapore (5%). The rest of the studies were conducted in 17 different countries, but with less frequency—one to two studies in each country. Supporting Information Appendix C contains a table of the nations where each of the included studies was conducted.

4.1.3. Publications by Year and Country. Many studies have been conducted in various countries during the past 5 years. The number of countries focusing on this research reached a peak in 2021. In 2018, one study was conducted in Israel [51]. In 2019, studies were conducted in the United States, Germany, and China. In 2020, studies were conducted in the United States, Germany, and Finland. In 2021, a jump in the number of countries that conducted XAI studies occurred. Studies were conducted in 13 countries: Sweden, Spain, Singapore, Portugal, Korea, Italy, Ireland, India, China, Australia, the United States, the United Kingdom, and the Netherlands. In 2022, studies were published in 14 different countries: Belgium, Canada, China, France, Germany, Ireland, Israel, Italy, Korea, the Netherlands, the United States, Singapore, Sweden, and the United Kingdom. In 2023, studies were conducted in Australia and Singapore.

4.1.4. Publication Type. As can be seen in Figure 4, among the 101 included studies, 48 were conference papers (47.5%) and 41 were journal articles (40.5%). Only 11 book section publications were included in this review (10.8%). The most frequent publication venues in Subsection 4.1.1 were conferences. A detailed table of the publication types of the included studies is shown in Supporting Information Appendix D.

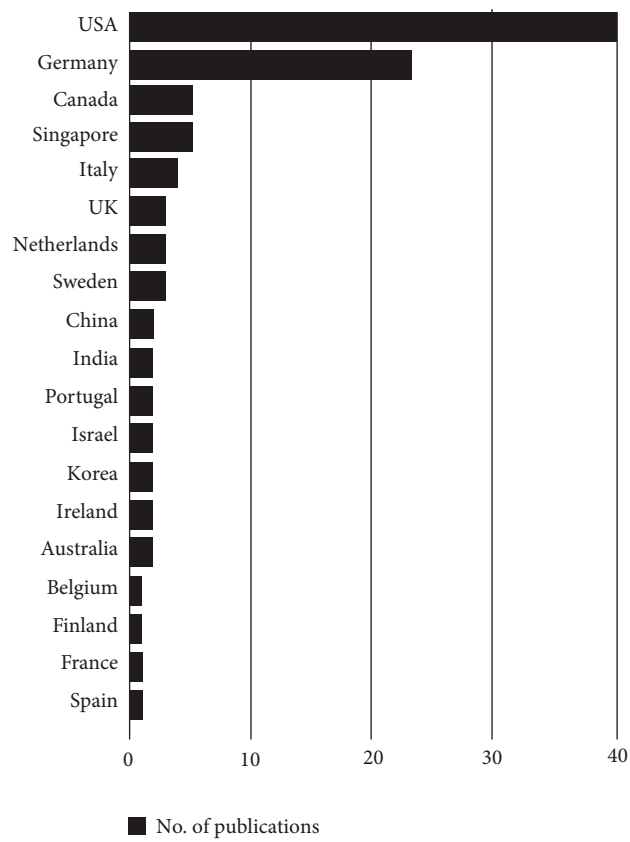

Figure 3: Number of publications in each nation.

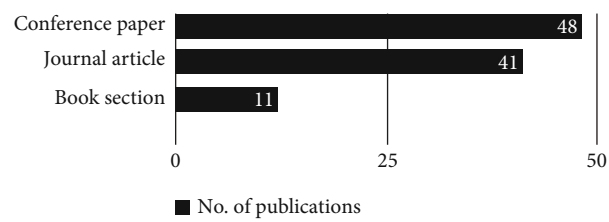

Figure 4: Publication type of included studies.

4.1.5. XAI Goals Addressed in Reviewed Domains. In this research, studies were categorized according to the XAI goals the authors aimed to achieve in different domains: trustworthiness, informativeness, interactivity, fairness, confidence, accessibility, transferability, and causality. As can be seen in Figure 5, the Sankey diagram shows that the most frequent XAI goals that researchers address in their work are trustworthiness (42.6%), informativeness (32.6%), and interactivity (10.9%). The frequency of the research conducted to achieve other XAI goals was very low comparatively, such as confidence (4.8%), fairness (8.5%), accessibility (2.4%), causality (1.2%), and transferability (2.4%). A detailed table of the selected studies aimed at achieving XAI goals is shown in Supporting Information Appendix E.

Achieving XAI goals has been demonstrated in a diverse range of domains, with a total of 23 fields identified and categorized into nine distinct domains. These domains include the medical domain ($n = 15$), economics ($n = 11$), education

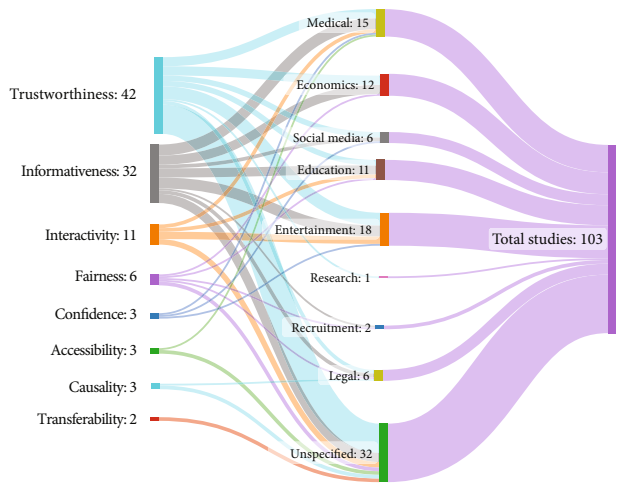

Figure 5: Sankey diagram showing the relationship between XAI goals and the domains of the included studies.

($n = 10$), research ($n = 1$), social media ($n = 6$), legal ($n = 6$), recruitment ($n = 2$), and entertainment ($n = 18$). Notably, 29% of the included studies ($n = 32$) were conducted in unspecified domains solely for the purpose of evaluating XAI systems. In Figure 5, the Sankey diagram shows the frequency of studies in various fields and the XAI goals addressed in these domains. For instance, to achieve trustworthiness, Malhi et al. [52] analyzed how human–agent interactions can lead to decision-making biases as users perceive and trust visual explanations differently in a loan application system. In terms of informativeness, Yu et al. [53] involved participants interacting with an XAI system that provides personalized advice. A detailed table is in Supporting Information Appendix F.

4.1.6. Types of ML Models Used to Achieve XAI Goals. In this subsection, ML models used to develop different AI systems were mapped to explore whether there is a correlation between XAI goals and the ML models used. Various types of ML and DL models and methods were used as tools to achieve XAI goals, as shown in Table 6. The most used ML model was a DL-based model, used in 40.5% of the included studies. Following this, classification models were employed in 34.5% of the studies, while regression models were used in only 14%. On the other hand, the natural language processing model (5%), the rule-based classifier model (2%), and the open learner model (1%) were some of the least commonly used models. Based on mapping XAI goals against ML models, it was concluded that they are uncorrelated.

Regarding DL models, four types of DL-based models were identified in the included studies. In 17 studies, convolutional neural networks (CNNs) were used to achieve various XAI goals, including trustworthiness [16, 22, 34, 54, 55, 66], transferability [18], accessibility [56], fairness [35, 56–58], causality, confidence, and informativeness. Deep neural networks (DNNs) were implemented in 13 studies to achieve informativeness, transferability, interactivity, and trustworthiness. Recurrent neural networks (RNNs) were utilized in five studies with the focus on achieving trustworthiness and informativeness in XAI systems. Additionally, six studies did not specify the type of neural networks (NNs) used, while two employed the attractiveness visual feature (AVF) model [66, 67] to address goals such as interactivity, informativeness, causality, interactivity, and trustworthiness.

Table 6: Types of machine learning models used to reach XAI goals.

ML model	No. of publications	Study ID
Deep learning-based models		$N = 41$
CNN	17	[16, 18, 19, 22, 34, 54–65]
DNN	13	[30, 63, 66–75]
RNN	3	[16, 63, 76]
NN	6	[20, 35, 77–80]
Classification models		$N = 35$
Artificial neural networks	4	[69, 81–83]
Decision trees	11	[35, 47, 52, 69, 77, 81, 84–88]
SVM	8	[35, 69, 81, 85, 86, 89–91]
Random forest classifier	8	[34, 69, 91]
Rule-based classifier	2	[92, 93]
Ensemble trees	1	[81]
Supervised classification	1	[94]
Text classifier	3	[95–97]
Regression models		$N = 15$
Logistic regression	9	[17, 70, 86, 98–103]
Linear regression model	6	[35, 69, 81, 104–106]
Natural language-based models		$N = 6$
Natural language processing	6	[93, 107–111]
Chatbot	2	[112, 113]
Explainable boosting machine	1	[114]
Open learner model	1	[115]
Recommender system	16	[21, 51–53, 116–127]

Among the seven classification models identified, ensemble trees, supervised classification, and text classifiers were each used once to achieve XAI goals, such as confidence [81] and informativeness [94–97, 107]. Decision trees (DTs) were used in nine studies to accomplish the XAI goals of informativeness, fairness, trustworthiness, and interactivity [35, 47, 48, 52, 68, 77, 84–88]. Support vector machines (SVMs) were used in eight of the included studies to address fairness, trustworthiness, interactivity, informativeness, and trustworthiness [35, 68, 69, 85, 86, 89–91]. Random forest classifiers were utilized in eight studies to achieve informativeness, fairness, and interactivity [69, 84–86, 128–131]. Finally, a rule-based classifier was employed in two studies

to target the XAI goals of informativeness and trustworthiness [30, 92].

In terms of other ML models employed in the selected studies, two types of regression models were employed. The first was logistic regression, which was used in multiple studies to achieve the goals of interactivity, fairness, trustworthiness, and informativeness [17, 70, 86, 98–103]. The second was the linear regression model, employed in several studies to attain the goals of trustworthiness, interactivity, informativeness, and confidence [35, 69, 81, 104–106]. Among the 101 included studies, six used a natural language-based model. All these studies are aimed at achieving trustworthiness, an XAI goal [107–110, 116, 117, 129].

Although 20 studies did not specify the ML models used, they still employed AI systems, such as recommender systems and chatbots, to evaluate various XAI goals. Recommender systems were used in 16 studies to address the goals of trustworthiness, informativeness, and interactivity [21, 51–53, 116–127]. Chatbots were employed in two studies to attain the XAI goals of informativeness and confidence [112, 113]. Other models, including the explainable boosting machine and the open learner model, were each used in one study to achieve the XAI goal of trustworthiness [114] and interactivity [115].

In conclusion, the diverse range of ML models used in the development of AI systems for XAI demonstrates the ongoing efforts to align XAI goals with appropriate techniques. Researchers can further enhance and progress the subject of explainability by identifying the connections and patterns between XAI aims and the ML models used, ultimately leading to the deployment of AI systems in diverse domains.

4.2. Explanation Methods. This work also examines the local and global explanation methods utilized in the XAI systems of various studies, as shown in Figure 6. Local explanation methods provide users with insight into the factors and variables that have the most significant impact on the AI system's prediction [132]. In contrast, global explanation methods enable users to understand how all of the features and variables in a ML or DL model contribute to its overall prediction [132]. According to Meske and Bunde [132], local explanations are generally preferred for novice users who may find it difficult to understand global explanations of an ML model.

4.2.1. Local Explainability. Interestingly, 43% of the studies included in this review implemented local explainability to interpret ML and DL models. This approach is especially useful in domains where the users are predominantly novice users and lack technical expertise [132]. In summary, understanding the local and global explanation methods of XAI systems is essential for users to gain insight into the AI model's predictions. While local explanations may be more suitable for novice users, global explanations can offer more comprehensive insights into the ML or DL model's workings [18, 20–22, 30, 40, 48, 51, 52, 54–61, 63, 65, 67–75, 78–80, 82, 84, 85, 90, 92, 96, 97, 100, 102–105, 107–110, 112, 116, 120, 121, 123, 128–136].

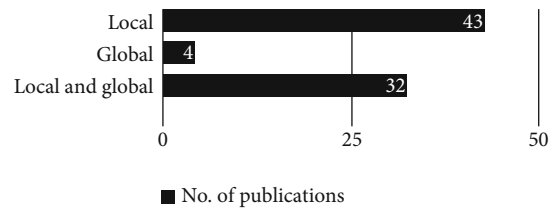

Figure 6: Local and global explanations of included studies.

4.2.2. Global Explainability. Global explainability was used in only 4 of the 101 studies analyzed. Kim et al. [16] focused on using visual explanations to rank motion-capturing activities, while Kumar et al. [34] developed an XAI system to analyze tumor images. In a third study, Liu et al. [105] developed an interactive AI system to support collaborative decision-making by involving a specialist in analyzing patient conditions and providing feedback to improve predictions. In the fourth study, Wang and Yin [17] examined four XAI methods for decision-making tasks utilizing global explainability and discovered that such explanations did not match the desires of individuals with less AI knowledge.

4.2.3. Local and Global Explainability. According to various experts, local and global explainability was used in 32 of the papers included in this review [19, 47, 62, 64, 66, 76, 83, 86, 88, 89, 93–95, 98, 99, 101, 106, 111, 113–115, 117–119, 122, 124–127, 137]. Figure 7 provides a visual representation of the frequency of studies that used both local and global explanations, indicating that a substantial portion of the studies utilized both methods. The integration of both local and global explainability methods can provide users with a more comprehensive understanding of how a ML or DL model works and can enhance the interpretability of the system.

It can be deduced that the use of local and global explainability methods in XAI systems can significantly improve the interpretability of ML and DL models. While local explanations are generally preferred for novice users [132], the integration of both local and global explanations can provide users with more comprehensive insights into the AI model's workings. However, the limited use of global explainability in the analyzed studies suggests a need for further exploration and development of this approach, particularly in areas where it could be most beneficial to users with the varying levels of AI expertise. A chi-square test of independence was conducted to examine the relationship between explanation methods and participants' expertise. There were five different categories of participants: domain experts, students, the general audience, crowdsourced individuals, and preadolescents. The frequency of studies of participants' expertise was categorized under explanation method categories, which are local explainability, global explainability, and local and global explainability. The relation between variables was not significant ($X^2(1, N = 101) = 5.32$, $p = 0.723$). The explanation methods employed to develop XAI systems do not depend on the expertise of the recruited participants. As the field of XAI continues to evolve, it is essential to prioritize the development and implementation

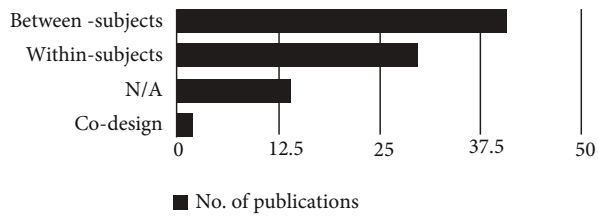

Figure 7: No. of publications for different study designs.

of both local and global explainability methods to enhance the interpretability and transparency of AI systems.

4.3. Detailed Procedure of Selected Studies. In order to have a thorough understanding of how explainability is studied in user-centered research, this article reviews the user studies in the included papers. This includes the research methods, study design, participants and their expertise, data collection methods, tools for evaluating XAI systems, evaluation mediums, and analysis methods. Most of the papers included have conducted experimental or user study research methods, while the use of mixed methods is rare. Crowdsourcing platforms were frequently used to recruit participants. As for data collection methods, questionnaires were the most common, whereas interviews and observational methods such as think-aloud protocol and recordings of video, audio, and mouse clicks were rarely used to collect data. Also, questionnaires and surveys were used as evaluation tools, and the results allowed researchers to draw conclusions regarding the XAI system presented to participants. These XAI systems were mostly web applications that participants in the selected studies were required to interact with, while graphic UIs were rarely used (3.9%). The majority of the included studies used a quantitative analysis approach to analyze the data collected throughout the study. The last part of this section identifies challenges that are either user-related, scope-related, or methodology-related. The following sections dive into each of the aspects described above separately.

4.3.1. Research Methods. A variety of research methods were employed in the 101 studies included in this review. An experimental study approach was implemented in 50 studies (49.5%), a user study approach was used in 44 studies (43.5%), and a mixed methods approach was used in 6 studies (7.4%). Although two studies did not explicitly specify the research method followed in their study, the methodologies employed enabled the researchers to conclude that those studies were user studies. All studies consisted of one or more conditions that were introduced as interventions to measure their effect on the participants while performing the task. These conditions either apply to the ML models used to generate the explanations or to the XAI representation methods. Overall, this review highlights the range of research methods employed in the studies and demonstrates how interventions are introduced by researchers and evaluated by users.

A detailed table of studies that employed different research methods is shown in Supporting Information Appendix G.

4.3.2. Study Design. Among the 101 studies analyzed, three study designs were identified: between-subject, within-subject, and codesign. Between-subject design was the most used (50.4%), followed by within-subject design (37%). Only two studies utilized the codesign approach, both conducted in the medical domain with medical experts. She et al. [64] used XAI to investigate post-treatment atrial fibrillation (AF) and held codesign sessions with doctors, whereas Wang et al. [136] presented a user-centered XAI system for clinicians as a diagnostic tool for critical care phenotyping. However, 13% of the studies did not specify the study design approach used. In summary, while between-subject and within-subject designs are the most commonly used approaches in XAI studies, the codesign approach has shown potential in the medical domain, highlighting the importance of tailoring study designs to the specific needs and expertise of the participants. A table displaying the frequency of each study design can be found in Figure 7.

4.3.3. Participants. This review explores the number of participants involved in the 101 studies and their expertise. The number of participants varied between studies, ranging from 3 to 3,500. In 42 studies, the sample size was fewer than 50 participants (41%). In 26 studies, the sample size was between 51 and 100 participants (25%). Only four studies had sample sizes greater than 900 (see Supporting Information Appendix H).

This systematic review is aimed at understanding to what extent it is challenging to recruit domain experts and how researchers substitute them. The reason is that the challenge of recruiting participants of varying levels of expertise may vary from one domain to another. The selected studies relied heavily on easily accessible populations, such as those recruited from crowdsourcing platforms ($n = 42$) and students ($n = 30$). The nonrepresentative sample from crowdsourcing platforms refers to the recruitment of easily accessible, undefined population groups who pass the attention check or the required approval rate. Some studies recruited general population participants ($n = 15$). Domain experts were only recruited in 18 studies. The studies have been categorized into various groups below, depending on the number of participants involved.

4.3.3.1. Studies Recruiting Fewer Than 50 Participants. Out of the 42 studies that recruited 50 participants or less, 18 studies recruited domain professionals such as critical care providers [84], experts in pathology and neuropathology [137], site reliability engineers (SREs) [113], legal professionals [19], researchers [116], professional fraud analysts [129], doctors [34], stroke rehabilitation therapists [35], professional game designers [72], cardiologists [64], annotators [114], senior residents and a medical officer [136], audio processing experts [131], employees with administrative positions [90], and AI experts [60]. Thirteen studies recruited university students [16, 63, 65, 75, 81, 88, 106, 108, 112, 119, 128, 132, 134], while one study recruited adolescents for an educational XAI system [125]. Ten studies recruited general population participants, such as individuals with ML knowledge [82], customers [61], and people

with technical and nontechnical backgrounds [109]. There were also several unspecified general population groups. Three studies recruited crowdsourced participants [80, 100, 123] (see Figure 8).

4.3.3.2. Studies Recruiting 51–100 Participants. The variety of recruited participants continued in studies that had 51–100 participants: 8 studies recruited participants from crowdsourcing platforms [56, 76, 81, 83, 115, 118, 126, 132], 5 recruited university students [52, 53, 74, 94, 98], 4 recruited participants identified as general audience members [18, 22, 73, 89], and 2 studies did not explicitly state the type of participants recruited [48, 92] (see Figure 8). Preadolescents were identified as a new group of participants in this category. Melsión et al. [57] aimed to improve preadolescent students' understanding of gender bias and discrimination in AI system predictions. The system explained its possibilities of bias and discrimination and highlighted how suitable the prediction system was for educational programs [57].

4.3.3.3. Studies Recruiting 101–200 Participants. Among the 101 studies selected for analysis, only 17 studies recruited 101–200 participants. Nine of these studies recruited participants from crowdsourcing platforms [54, 67, 89, 99, 102, 122, 124, 128, 132], six recruited university students [20, 30, 51, 93, 104, 133], and two recruited general population members [95, 131] (see Figure 8). The variety of participants started to decrease as the number of participants increased, as shown in Figure 8.

4.3.3.4. Studies Recruiting More than 200 Participants. Fourteen studies recruited more than 200 participants. Eleven studies recruited participants from crowdsourcing platforms [17, 55, 62, 69, 81, 91, 105, 117, 121, 132, 135]. Two studies recruited university students [51, 93], whereas one study recruited participants identified as general audience members [71].

In summary, this review has identified several domains where XAI systems are being targeted towards domain experts, such as the medical domain ($n = 11$), economics ($n = 12$), and the legal domain ($n = 8$). A Pearson correlation coefficient was computed to assess the relationship between the domain and frequency of studies in relation to the number of participants. This test focused on domains that had a high frequency of studies and recruited 50 participants or less, primarily domain experts. The correlation analysis revealed a negative correlation between the domain and the number of participants, indicating that as the number of participants increased, the frequency of studies conducted in these domains decreased. The statistical analysis for the two variables revealed r(13.7) =6 and p = (0.03). So, this indicates that researchers face significant challenges regarding the recruitment of a large number of domain experts to have reliable data that is sufficient for analysis. Thus, recruiting participants via readily accessible crowdsourcing platforms and managing experiments and user studies is much more feasible.

4.3.4. Data Collection Methods Used in the Included Studies. This review examines the design and methodology employed in the analysis of 101 selected studies, focusing

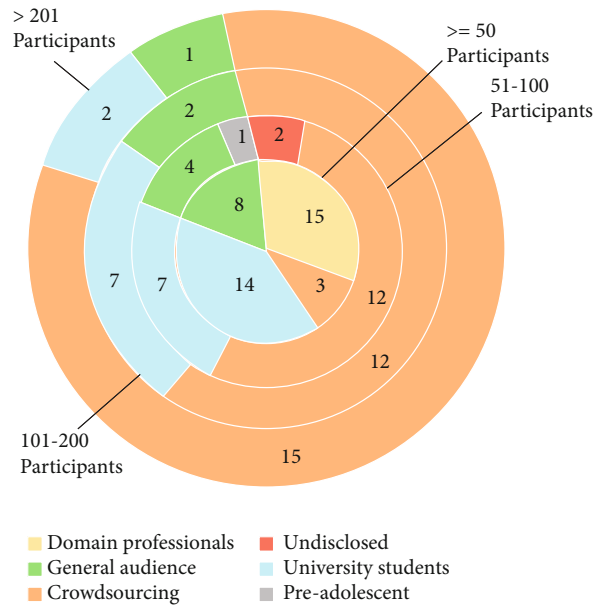

Figure 8: Studies classified to categories based on the number of participants.

on their respective objectives. The studies utilized various data collection methods, including questionnaires, surveys, interviews, and observational techniques. Notably, a majority of the studies employed multiple data collection methods, demonstrating a preference for comprehensive approaches rather than relying solely on a single method. Furthermore, these methods were employed both prior to and after the primary study task, enabling a comprehensive exploration of the research questions at hand. A detailed table is shown in Supporting Information Appendix I.

4.3.4.1. Surveys and Questionnaires. Sixty-one percent of the included studies used surveys or questionnaires for data collection. Researchers in 40% of the included studies have used validated scales not only to understand their participants but also to evaluate the achievement of different XAI goals. On the other hand, in 21% of the studies, researchers have tailored questionnaires and surveys to suit their objectives.

Validated surveys and questionnaires have been used in several studies to evaluate the achievement of XAI goals. For example, participants filled out the Propensity to Trust questionnaire before the research and the Trust in Automation questionnaire after the study to explore whether explanations in a trip planning assistance system increased trust [119]. Anderson et al. [20], for instance, conducted a study in the gaming industry that used a questionnaire to measure cognitive load and assessed the informativeness of the XAI system [20]. In a study conducted in the legal domain to develop an interactive decision support XAI system, Liu et al. [105] used a survey to measure the usefulness and richness of the system to support decision-making.

In terms of tailored surveys and questionnaires, several studies used surveys to gather participants' demographics [93, 99, 133], assess knowledge, and understand preferences and for many other purposes. For instance, Miranda et al.

[72] conducted a study in the gaming domain and ran a questionnaire before the task to understand and evaluate participants' skills using the Pac-Man video game. This was because control would be given to a human player if the AI system reached an unknown state in the game. The system would then learn from the human player and explain its behavior in future moves [72]. Musto et al. [93] provided a questionnaire to the participants to understand their book and movie preferences before they began to use an explainable recommender system. Some prestudy surveys were given to participants to understand their background knowledge in various fields [30, 69] and questionnaires to understand their personalities, such as the Big Five Traits questionnaire [51, 114]. Hernandez-Bocanegra and Ziegler [122] provided a quality check of participant responses at the beginning of the study to decide whether they were eligible to proceed [122]. In their work, Kenny et al. [54] developed three experiments and provided a survey to participants before the beginning of each experiment [54]. In a study that targeted children to help them understand gender bias in AI, Melsión et al. [57] tailored two questionnaires for the control and experimental groups before proceeding with the given task.

In this review, 52% of the studies used poststudy surveys and questionnaires. For example, several studies provided participants with the Trust in Automation questionnaire to measure the users' levels of trustworthiness in the XAI system [68, 82, 119]. Other studies measured different aspects; for example, Berger and Müller [92] measured user impressions in a study focused on gaming; Chromik and Butz [26] measured self-ratings of perceived understanding and objective understanding for a loan application XAI system; and Sovrano and Vitali [83] provided their participants with a system usability questionnaire to evaluate a loan decision support system. In a survey that measures experiences of frustration, Wolterink and Bakkes [131] aimed to achieve interactivity as an XAI goal in a Pac-Man game that developed an explainable frustration prediction system [131]. However, Abdulrahman et al. [98] gave participants the questionnaire as part of the study rather than before or after the study. The questionnaire was provided to measure stress while interacting with an interactive, explainable stress prediction system [98].

4.3.4.2. Interviews. Interviews were carried out in 14% of the studies as a data collection method. Among these studies, six conducted semistructured interviews [40, 88, 112, 113, 132, 137]. For instance, in a study focused on hate speech on social media, Meske and Bunde [132] developed explainable decision support systems and conducted a semistructured interview with social media moderators to assess their perception of the system in the first stage of evaluation. In another study, Wang and An [65] conducted an in-depth interview with 3 out of 30 participants to understand their opinions about an XAI image recognition system [65].

4.3.4.3. Observational Data Collection Methods. Some observational data collection methods were identified in some of the included studies. For example, a think-aloud protocol occurred in six studies to understand the participants' train of thought while using the system [69, 76, 80, 105, 111, 128]. In a study focused on loan application, Chromik et al. [128] conducted a think-aloud protocol with 15% of the participants to understand their train of thought while using the system. Another technique, used in 11% of the studies, was to collect audio/video recordings and mouse clicks to analyze the impact that XAI has on participants [16, 20, 40, 48, 54, 80, 84, 88, 111, 125, 132]. Ooge et al. [125] collected mouse clicks from adolescents in a system that demonstrated the importance of personalized recommendations and clear explanations to improve user engagement, understanding, and trust in e-learning. Prange et al. [40] conducted a study aiming to develop AI systems that aid medical practitioners in diagnosing and treating neurological conditions. Specifically, the researchers implemented real-time tracking of input from elderly participants undergoing dementia testing [40].

This review explored the relationship between data collection methods and their alignment with XAI goals, domains where the selected studies were conducted, and participant expertise. The results of the statistical analysis showed no significant association between the most frequent XAI goals, namely, trustworthiness, informativeness, and interactivity, and the employed data collection methods ($X^2(1, N = 94) = 7.8, p > 0.05$). Similarly, no significant association was observed in the data collection methods used across various domains ($X^2(1, N = 144) = 15.7,\ p > 0.05$). However, the chi-square test of independence indicated a significant relationship between data collection methods and participants' expertise ($X^2(1, N = 98) = 26.9, p = 0.0014$). This emphasizes the significance of considering user characteristics in formulating data collection strategies for the effective implementation of XAI systems. The results did not establish a direct influence of XAI goals or domains on data collection methods. However, the results above shed light on the equitable nature of data collection techniques across diverse domains, thus contributing valuable insights to the field of XAI research and practice.

4.3.5. Evaluation Tools. Various evaluation tools were given to participants to assess the XAI systems developed in the studies included in this review, as shown in Table 7. More than half of the included studies (63%) used online surveys or questionnaires as evaluation tools. Examples of information gathered in the studies include ratings of the relevance of explained recommendations in a dating recommender system [51]; the influence of explained attributes on customer satisfaction in a competitor analysis system [61]; open-ended questions given to professional game designers to describe their thoughts when using the system as a tool to train the Pac-Man character [72]; and levels of confidence [62], rating acceptance of recommendations [117], and ranking explanations [103]. In addition, four studies interviewed participants to understand how they perceived the system [64, 86, 113, 132]. Two studies asked the participants to develop an affinity diagram to brainstorm and evaluate the proposed XAI systems [65, 136].

Table 7: Evaluation tools used in the included studies.

Evaluation tool	No. of publications	Study ID
Survey/questionnaire	68	[17–19, 21, 22, 30, 34, 35, 40, 51–53, 55, 57–64, 68–70, 72–75, 77–81, 83, 87–91, 93, 95–98, 101–110, 112, 114, 117, 120, 123–128, 130, 131, 134, 135] [2–4, 14, 17, 18, 23, 24, 28, 37, 42, 43, 47–49, 55, 58, 63, 64, 66, 68, 70–73, 75, 77–82, 86–88, 90–94, 98, 102–107, 110, 112–115, 117–121, 125–127, 130, 131, 133–137]
Interview	4	[64, 86, 113, 132]
Affinity diagram		[65, 136]

Table 8: Medium of evaluation used in the included studies.

Medium of evaluation	No. of publications	Study ID
Web application	34	[17, 18, 21, 35, 51, 55, 57, 60, 62, 73, 74, 78, 79, 81, 87, 89, 93, 95, 96, 98, 104, 105, 108–110, 114, 117, 124–127, 130, 134, 138]
Interactive interface	4	[52, 64, 74, 98]
User interface (UI)	32	[19, 20, 30, 34, 48, 53, 58, 59, 63, 65, 68, 70, 72, 75, 77, 80, 83, 86, 90, 91, 94, 101, 102, 106, 112, 113, 116, 128, 131, 135, 136]

4.3.6. Medium of Evaluation. This review identified four categories of mediums used for evaluating the explanations of AI systems. The first category is web applications, which were used in 33% of the included studies, followed by UIs (31%) and interactive interfaces (4%). A detailed list of studies and the medium of evaluation used is shown in Table 8.

4.3.7. Analysis Methods. The studies included in this systematic literature review used quantitative and qualitative approaches to analyze the data collected, as shown in Figure 9 and Supporting Information Appendix J. The quantitative data analysis approach was followed in 45 studies, and the qualitative approach was present in 5. Twenty-nine studies used a combination of quantitative and qualitative approaches.

4.4. Challenges. After exploring the challenges and limitations that researchers identified in the studies, three distinct categories were established to organize these challenges, namely, user-related, scope-related, and methodology-related challenges, as shown in Figure 10. These categories provide a comprehensive framework for understanding and addressing the various obstacles encountered during the evaluation of XAI systems. This section also sheds the light on studies lacking explicit identification of challenges. User-related challenges encompass issues related to the participants or end users involved in the studies, such as their diverse backgrounds, expertise levels, and cognitive abilities. Scope-related challenges pertain to the broader contextual factors that influence the evaluation process, including the complexity of the XAI system, the specific application domain, and the scalability of the evaluation methods. Methodology-related challenges focus on the specific techniques, tools, and approaches employed in the evaluation, including data collection, analysis, and explanation methods. The categorization of these challenges into three distinct categories provides researchers and practitioners with a clearer understanding of the potential hurdles they may encounter when evaluating XAI systems, allowing for more targeted and effective strategies to overcome these challenges.

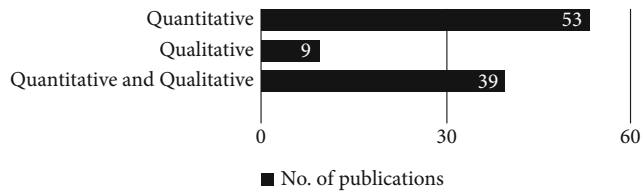

Figure 9: Analysis methods of the included studies.

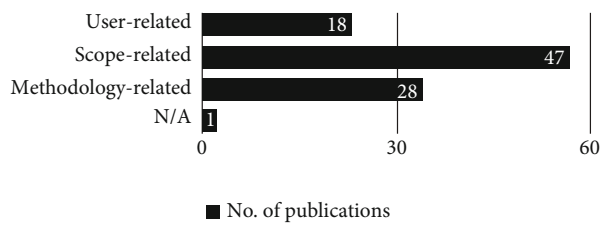

Figure 10: Challenges identified in the included studies.

4.4.1. User-Related Challenges. User-related challenges were identified in 17% of the included studies. These challenges encompassed a range of issues related to the participants or end users involved in the evaluation process. Examples of user-related challenges include instances where the sample size was limited, hindering the generalizability and statistical power of the findings [88, 100, 125, 137]. In a study conducted by Flutura et al. [82], participants had varying levels of knowledge about ML; more efforts were required to enhance their understanding of the system to be evaluated [82]. Difficulties were also observed in terms of users comprehending the visual representations of XAI for AI models in studies conducted by Górski et al. [19] and Gupta et al. [47]. Furthermore, user expectations played a role in some studies; for instance, participants felt overwhelmed or

dissatisfied when asked too many questions during the recommendation process, resulting in a system that did not meet their expectations [126]. These user-related challenges shed light on the complexities involved in effectively engaging participants, ensuring their understanding and satisfaction, and tailoring the XAI systems to their needs.

4.4.2. Scope-Related Challenges. Scope-related challenges were identified in a significant proportion of the studies included in this review, approximately 46.5%. These challenges primarily revolved around the limitations or narrow focus within the scope of the studies, indicating potential missed opportunities for broader insights and more comprehensive evaluations. For instance, Alqaraawi et al. [18] utilized a small dataset that restricted the examination to a single technique for generating saliency maps, whereas the inclusion of multiple techniques could have offered a more robust and comparative analysis [18]. This limitation not only restricted the diversity of approaches but also potentially limited the depth and breadth of the findings. Likewise, Lee et al. [35] suggested that broadening the scope to obtain more data could have provided better insights, leading to better results. Only two types of explanations were taken into consideration in a study by Conati et al. [94], which may have overlooked other valuable explanatory methods or models that could have enriched the evaluation and provided a more comprehensive understanding [94].

4.4.3. Methodology-Related Challenges. Methodology-related challenges were found in 27% of the studies reviewed, indicating issues related to the specific methods employed for data collection or evaluation. These challenges hindered the smooth execution or comprehensive analysis of the study. For instance, a study conducted by Prange et al. [40] encountered technical difficulties when attempting to record the movements of a digital pen that study participants were using. This issue highlights the importance of minimizing potential disruptions, as technical glitches or limitations can impact the quality and validity of the results. In another study, Guesmi et al. [116] identified the need for more extensive quantitative and qualitative data to fully explore the correlation between user characteristics and the level of detail required in the explanation [116]. These methodology-related challenges highlight the importance of having well-defined methodologies in XAI evaluation studies. By carefully selecting and implementing appropriate data collection and evaluation methods, researchers can enhance the reliability, validity, and comprehensiveness of their findings.

4.4.4. Studies Lacking Explicit Identification of Challenges. In the analysis of the included studies, it was observed that despite the presence of challenges or limitations in almost all the studies, 6% has not explicitly identified or discussed the difficulties, limitations, or obstacles encountered during the conducted research [58, 59, 65, 77, 106, 109, 110]. One example is the study conducted by Wang and An [65], who developed an interactive mobile application to aid users unfamiliar with CNNs. The study assigned users the task of capturing photos, with the XAI application revealing the model's decision-making process [65]. Although innovative in supporting nontechnical users, the study did not explicitly discuss encountered challenges or limitations.

Overall, this review highlights the significance of recognizing and addressing user-related, scope-related, and methodology-related challenges in evaluating XAI systems. By addressing these issues, researchers can enhance the development, implementation, and effectiveness of XAI systems, ultimately benefiting end users and advancing the field of XAI.

5. Discussion

The discussion focuses on three observations from this review. First, this section discusses the choice of data collection methods used in the included studies. The choice of data collection methods is driven by the domain where the research is conducted, as well as the XAI goal that researchers aim at achieving. This section also sheds light on some methods that were rarely used in the included papers, which might be advantageous if used in other studies. Second, it reviews participant recruitment, including understudied populations. Third, this section discusses the choice of XAI representation methods and how it is correlated to user expertise and the data utilized to provide the explanation. Overall, the Discussion section is concluded by introducing six pillars that prove the interrelationship between the domain explored in a study, the intended XAI goal, the data collection method, the targeted population, explanation, and representation XAI methods. These pillars could be followed when implementing a user-centered study to select appropriate data collection approaches that allow researchers to effectively evaluate XAI systems.

5.1. Choice of Data Collection Methods. The analysis of this review highlights three key observations regarding the choice of data collection methods. First, the choice of data collection methods is influenced by the domain addressed in the study. Second, there are underused data collection methods that have the potential to offer valuable insights to researchers. Third, the choice of data collection methods is also guided by the XAI goal pursued by researchers in their work. Understanding the domain, exploring underused data collection methods, and aligning with the specific XAI goal are essential considerations for researchers when selecting data collection methods to evaluate XAI systems.

5.1.1. Data Collection Methods Driven by Domain. One of the goals of this review is to identify the contribution of HCI evaluation techniques to achieving XAI goals. Overall, the choice of HCI evaluation techniques used to collect data and evaluate XAI systems often depends on the domain and the XAI goal to be achieved. When selecting a method, researchers often tradeoff the data collection methods they might consider using in their study design. This review sheds light on the tradeoffs to provide researchers with a clear understanding of their choices in future research. Despite the fact that the statistical analysis in Section 4.3.4 showed

insignificant results, the medical domain serves as a critical example where decision-making carries significant consequences. In such cases, researchers strive to maximize information gain to better understand end users, who may be medical professionals, including doctors [34], cardiologists [64], therapists [35], or medical residents [136]. To ensure the acceptance and adoption of XAI systems in this domain, studies conducted interviews as a data collection method, often supplementing it with other approaches such as pre-task and post-task surveys. By incorporating multiple methods, researchers aimed to gather comprehensive insights from medical professionals, enhancing the richness and depth of the data obtained. For instance, She et al. [64] explored explainability for post-treatment AF and performed in-depth interviews with the participants before and after completing the task.

Training sessions were implemented prior to the main task in domains where the intended users were experts in their respective fields. This practice is aimed at ensuring the appropriate utilization and evaluation of the XAI systems. In the medical domain, several studies incorporated training sessions for medical experts before conducting the main study [35, 40, 64, 88]. Similarly, in the educational domain, Conati et al. conducted a study where computer science students underwent a training session to familiarize themselves with the functionalities of the XAI system [94]. Additionally, four other studies in the educational domain followed a similar approach of providing training prior to the experiment [57, 86, 112, 127]. This approach ensured that the targeted users were adequately prepared and empowered to engage with the XAI system in the subsequent main task, enabling them to make informed decisions and derive meaningful insights from the system's explanations. In four studies conducted in the economics domain, training was conducted with students and crowdsourced participants [24], students [134], customers [61], and a general audience [83]. Looking at the instances mentioned at the beginning of the paragraph, it could be concluded that some participants were either domain experts who needed to be trained to use XAI systems to make complex decisions [86, 94, 116, 127] or lay users who were new to XAI systems [17, 57, 105, 112].

Surveys and questionnaires emerged as commonly utilized tools for data collection across various domains, indicating their versatility and flexibility in gathering information. Sixty-one percent of the analyzed studies used surveys or questionnaires that are either validated or specially tailored for the studies. Validated questionnaires were utilized in the reviewed studies to assess various factors, such as cognitive load [20], perceived usability, understandability, and system causality [87]. Additionally, domain-specific surveys were employed, such as the one developed by Miranda et al. [72] to evaluate participants' video gaming skills prior to engaging with a gaming XAI system. In the entertainment domain, five out of the 18 studies focused on the development of movie recommendation XAI systems. These studies implemented surveys and questionnaires before and after the task to gather participant feedback and insights [21, 101, 102, 124, 126]. These methods allow researchers to gather both quantitative and qualitative data, providing a comprehensive understanding of participants' perspectives, attitudes, and experiences related to the XAI systems under evaluation. By employing surveys and questionnaires, researchers can efficiently capture a wide range of insights and opinions, making them a valuable resource in user-centered evaluations of XAI systems.

Observational methods have been employed in several studies, such as the use of a think-aloud approach, recoding audio and video, and mouse clicks. However, in the context of this review, a definitive conclusion regarding the selection of observational methods as a preferred data collection method has not been reached. These methods were employed in different domains such as gaming [76], economics [26], legal [105], entertainment [80], and the medical domain [69]. While observational methods offer the advantage of capturing real-time user behavior and providing rich qualitative insights, their specific benefits and drawbacks in the context of XAI evaluations remain open to further investigation. One key consideration when employing observational methods is the goal of maximizing information gain. By directly observing participants' interactions with XAI systems, researchers can gain valuable insights into their decision-making processes, preferences, and challenges encountered during system usage. This can lead to a deeper understanding of the strengths and limitations of XAI models and provide valuable input for system improvements. However, the choice to use observational methods must also consider potential biases, as participants may alter their behavior when they are aware of being observed. This can impact the ecological validity of the study and introduce limitations in capturing natural user interactions. Given the complex nature of XAI systems and the diversity of evaluation contexts, the choice of data collection methods, including observational methods, should be carefully considered based on the research goals, study design, and specific requirements of the domain. Further research is needed to systematically examine the benefits and limitations of observational methods in different XAI evaluation scenarios and provide evidence-based guidelines for their optimal usage.

To conclude, interviews were commonly used in domains that develop XAI systems, targeting domain experts, to gather comprehensive insights from professionals. In terms of training given before the study, it was targeted towards participants in domains such as the medical, legal, economics, and educational domains. Participants who underwent training were domain experts in critical domains that required training to utilize XAI systems for decision-making, as well as novice users with no prior experience using XAI systems. Furthermore, surveys and questionnaires emerged as highly flexible data collection methods utilized across various domains, demonstrating their widespread applicability in research studies. Finally, observational methods, such as think-aloud approaches, audio and video recordings, and mouse click tracking, have been used in various studies to capture real-time user behavior and provide qualitative insights.

5.1.2. Underused Data Collection Methods. Upon reviewing Section 5.1.1, it becomes evident that interviews were rarely

utilized as a data collection method in the analyzed studies. It seems preferable for researchers to interview the targeted audience, whether domain experts [113] or students [112] rather than interviewing participants from a crowdsourcing platform. Out of the studies that recruited participants from crowdsourcing platforms, only 4% of them conducted interviews, accounting for 42% of the total studies. Six studies in the medical domain have conducted interviews with their participants, who are identified as medical experts [34, 35, 40, 64, 136, 137]. For instance, Lee et al. [35] examine the impact of an interactive XAI system that supports therapists in collaboratively reaching a decision. A poststudy interview was conducted after performing the task to understand participants' perspectives regarding the efficacy of the system and the influence its analysis on the therapists' decisions [35]. On the other hand, other domains such as entertainment, social media, education, and mere evaluation of XAI systems have conducted interviews in one study per domain [65, 112, 113, 132]. One takeaway from this analysis is that while interviews were not extensively used overall, they were more prevalent in studies involving domain experts.

5.1.3. Data Collection Methods Driven by XAI Goals. In the 101 analyzed studies, researchers combined data collection methods in a creative manner to gather valuable insights with regard to evaluating the achievement of the XAI goals defined by Barredo Arrieta et al. [8]. These data collection methods include questionnaires, surveys, interviews, and various observational methods, which are reviewed in detail in Section 4.3.4. Despite the nonsignificant outcomes of the statistical analysis conducted to evaluate the correlation between the utilization of specific data collection methods for the assessment of different XAI goals, several observations were uncovered in the process of data synthesis. In terms of the use of surveys and questionnaires, it is evident that the selection of a particular questionnaire or survey primarily depends on the specific XAI goal to be addressed in the study. As for interviews, the discussion in Section 5.1.2 showed that participants who are the targeted audience in the addressed domains are often interviewed. However, the case is different for observational techniques employed in the analyzed studies, which are discussed later in this section.

Different types of questionnaires to evaluate XAI goals were used to measure trustworthiness, informativeness, interactivity, etc. For instance, to evaluate trustworthiness, Bayer et al. [133] provided participants with a trust questionnaire to understand how user expertise may affect the level of trust in XAI systems. Several studies use the Trust in Automation questionnaire to measure trust in visual explanations in AI systems [48, 68]. In terms of evaluating informativeness, questionnaires such as perceived goodness of explanation [69] and perceived understandability [110, 120] were administered to participants after they performed the task to understand their perception of the XAI system.

Additionally, surveys and questionnaires were utilized to evaluate interactivity in a subset of the included studies (33% of 12 studies). These methods measured different aspects of the explanation such as its usefulness and richness to support decision-making [105]; a survey of how participants feel about the system [87]; and impressions of XAI methods, mental models, trust, computer efficacy, and cognitive workload [59].

It is important to note that studies addressing the rest of the XAI goals did not employ surveys and questionnaires that were specific to the goal. For example, Lee [61] used a survey that measured the influence the explainable attributes had on a customer's satisfaction with the recommendation [61]. Another study provided a survey to measure the perceived helpfulness and truthfulness of class activation mapping (CAM), which is an XAI representation method in the form of heatmaps [73]. The observation above reveals that trustworthiness, informativeness, and interactivity are among the top XAI goals investigated by researchers. Consequently, there is a substantial body of prevalidated measures available for evaluating these goals. However, for the less-addressed XAI goals, few researchers have developed validated measures that can be utilized in future studies. This indicates an opportunity for researchers to contribute to the field by developing and validating questionnaires and surveys specifically tailored to assess the remaining XAI goals. By doing so, they can enhance the methodological rigor and consistency of evaluating these underexplored aspects of XAI systems.

Observational data collection methods can also be valuable tools to collect data and evaluate XAI systems. Most studies employing observational techniques were conducted in the top three XAI goals addressed in the field of XAI, which are trustworthiness, informativeness, and interactivity. Eight out of the 46% of studies conducted to achieve trustworthiness have used observational techniques such as think-aloud protocol and recordings of audio, video, and mouse clicks [16, 40, 48, 54, 76, 105, 125, 132]. While four addressed informativeness [20, 84, 128, 132], one addressed interactivity [105], and one addressed accessibility [136]. Further exploration is required to better understand the reasons behind the relatively low usage of observational methods in the evaluation of XAI systems. While these methods have proven valuable in addressing certain XAI goals, such as trustworthiness, informativeness, and interactivity, their overall utilization remains limited.

5.2. Recruitment of Participants. An observation from the analysis of the studies is the researchers' preference for recruiting participants from easily accessible populations (i.e., participants recruited from crowdsourcing platforms, general audience, and students). Seventy-two percent of the 101 studies included in the analysis recruited participants from these populations. While the rationale behind this choice is apparent from a researcher's perspective, there is value in involving participants who are the actual target audience of the studied systems. This section discusses the importance of considering the inclusion of understudied populations to gather insightful data instead of relying solely on crowd workers in this field of research. By including a more diverse range of participants, researchers can gain a deeper understanding of the perspectives and experiences of the end users of XAI systems.

5.2.1. Understudied Populations. Within the subset of the 101 analyzed articles, a limited number of studies recruited individuals who are identified as domain experts despite the wide range of domains explored in this review, which are targeted towards domain experts (17% of studies). It is also important to note out that among the studies that recruited domain professionals, the number of participants did not exceed 33 [90]. This suggests that the avoidance of recruiting domain professionals may stem from the challenge of attaining a sufficiently large participant sample size to achieve statistically significant results. For example, in the domain of economics, Malhi et al. [52] developed a loan approval XAI system intended for individuals with expertise; however, they chose to recruit 65 university students instead. The choice to opt for nonexpert participants in certain domains may be driven by practical considerations and the difficulty of obtaining a sizable pool of domain experts for research purposes.

5.3. XAI Representation Methods. This review observed two XAI representation methods from the analyzed studies. The first method was the visual representation method, where heatmaps such as CAMs were used to explain the system's prediction. The second method combines textual and visual representation approaches, incorporating graphs, colors, highlighted text, and charts to enhance the textual explanations. This section reviews how XAI representation methods are driven by the data type being analyzed as well as the expertise of the participants involved in the studies.

5.3.1. Data Type Drives XAI Representation Methods

5.3.1.1. Visual Representation. Of the 101 studies included in this review, only 4% used CAM to explain the system's prediction. For example, Kumar et al. [34] used a CAM explanation of MRIs utilized in a brain tumor diagnostic system evaluated by 10 doctors. The analysis of MRI images requires image processing methods. Melsión et al. [57] used a CAM explanation on images shown to preadolescents to assess the fairness of the prediction system that applies image processing on the data [57]. Wang and An [65] examined a CAM explanation for an image recognition system evaluated by Amazon Mechanical Turk (AMT) participants [65]. Moreover, Zhang et al. [58] used CAM to investigate the system's fairness and ensure that it provided debiased predictions of processed images [58].

5.3.1.2. Visual and Textual Representation. Almost all the reviewed studies (96%) used visual and textual XAI representation methods derived from nominal and ordinal data. This utilization of visual and textual representation techniques showcases the diverse approaches taken to enhance the interpretability and transparency of the user-centered experience of XAI systems in the context of different domains such as the medical domain, economics, education, research, social media, legal, recruitment, and entertainment domains. One example of this approach is the development of an explainable movie recommendation system that incorporates textual explanations. Within these explanations, specific parts of the text were strategically highlighted using different colors to emphasize their significance [51].

5.3.2. User Expertise Drives XAI Explanation Methods. In this review, studies utilized local explanation methods, global explanation methods, and both. It is important to highlight that 60% of the included studies used local explanation methods, with a focus on nonexpert users (75%). The employment of local explanation methods in more than half of the studies aligns with the point explained in Subsection 2.1.1 that local explanations are often implemented in systems targeting lay users to ensure understandability. For instance, Musto et al. [93] developed a recommender system that uses users' book, music, and movie reviews to present local explanations using three forms of NL [93]. First, participants were asked to build their profiles by choosing their most preferred items among different domains. Then, the system provided an explained recommendation, and users were asked to rate it [93]. As for the rest of the studies, only 4% of the analyzed studies used global explanations alone.

5.3.2.1. Variety of Expertise and Recruitment of Participants. Looking at participant data in Subsection 4.3.3, it was observed that although the number of participants in studies continued to increase, the variety of participants started to drop, with only three categories of participants in later studies. For example, in studies that recruited 50 participants or fewer, there were four categories: domain professionals, university students, general audience members, and participants recruited through crowdsourcing platforms. Interestingly, the recruitment of domain professionals did not occur in any study with more than 50 participants (17% of the included studies). This observation supports the conclusion that, unlike other participant categories, recruiting a high number of domain professionals is challenging. In studies that recruited 51–100 participants, the participant categories identified were individuals recruited through crowdsourcing platforms (12% of included studies), university students (6% of included studies), students who were identified as preadolescents (1% of included studies), and general audience members (6% of included studies). Several studies did not specify the targeted audience or their expertise (2% of the included studies). From 101 participants onwards, the categories of participants in the included studies were individuals recruited from crowdsourcing platforms (26% of included studies), university students (9% of included studies), and general audience members (4% of included studies).

6. Pillars of User-Centered Evaluation of XAI Systems

To analyze researchers' selection of HCI techniques for evaluating the achievement of XAI goals in different domains, several factors need to be understood. These factors include the targeted domain, the intended XAI goal, the targeted population, the data collection methods, and the explanation and representation methods used to evaluate XAI goal achievement. However, in most reviewed papers, the

connections between these factors were unclear. This review emphasizes the importance of considering these factors collectively, as they are interrelated.

To ensure a coherent approach to evaluating XAI goals, it is crucial to identify the following:

1. Domain of the study.
   - o. Identify the specific domain of the study.
   - o. Focus on addressing the targeted issue within that domain using the XAI system.
   - o. Align efforts towards achieving the intended XAI goal within the domain.
2. XAI goal.
   - o. Define the specific XAI goal being addressed.
   - o. Identify suitable prevalidated questionnaires and surveys for participants. Trustworthiness, informativeness, and interactivity are commonly addressed XAI goals that already have prevalidated measures available.
   - o. Develop and validate new measures for XAI goals without prevalidated measures.
3. Targeted population.
   - o. Determine the targeted audience and assess their feasibility of recruitment.
   - o. Researchers could conduct interviews and use prevalidated measures for domain expert participants, as it provides valuable insights into their experience with the XAI system.
   - o. Consider lay users for studies targeting more general domains such as social media and entertainment.
4. Data collection methods.
   - o. The number of participants and their expertise affect the methods that are applicable for data collection in a study. For instance, it is feasible to conduct interviews with a few domain experts. This may not be achievable with a large number of participants recruited from a crowdsourcing platform.
5. XAI explanation methods.
   - o. Choose XAI methods based on the intended audience (local explanations for novice users and global explanations for AI experts).
6. Representation methods.
   - o. Select appropriate XAI representation methods based on the type of data used (visual explanations for image predictions and textual explanations with visual elements for nominal and ordinal data).

When considering the domain of study, XAI goal, targeted population, XAI explanation methods, and XAI representation methods, it is crucial to understand that these pillars are not isolated entities. They are interconnected and mutually influence one another, as shown in Figure 11. For example, the choice of domain influences the selection of an appropriate XAI goal and the explanation methods to be employed. Similarly, the targeted population impacts the selection of XAI explanation methods, data collection methods, and representation techniques. This allows researchers to make more informed decisions regarding the choice of data collection methods. They can ensure that the selected methods align with the specific goals of the study, the characteristics of the targeted population, and the nature of the XAI system being evaluated.

7. Strengths and Limitations

By conducting this systematic literature review, valuable insights have been gained regarding the evaluation of XAI systems across various domains, thus addressing a gap in the existing literature. However, it is important to acknowledge the limitations of this review, including the databases explored, the language of publications considered, and the selection of studies that clearly identify the ML models utilized only. These strengths and limitations are discussed in the following paragraphs to provide a comprehensive understanding of the scope and implications of this review.

7.1. Strengths. This review was conducted to help fill the gap created by the lack of systematic literature reviews covering all the domains where XAI systems are being developed. Other literature has only focused on one domain and reviewed how XAI was implemented. This review explored literature in various domains, such as medical, economics, education, social media, legal, and recruitment, and the user-centered procedures used to evaluate the XAI systems. This review is the first to provide a holistic view of the user-centered design approaches used to assess XAI in various domains. In addition, since this review only had one author that selects the studies to be included (N.H.A.A.-A.), a Cohen kappa test was conducted with two experts (D.A.A.-T. and R.S.A.-M.) applying the selection criteria to 20% of the included studies. The results of this test showed more than 80.8% agreement on the inclusion and exclusion of the retrieved studies.

7.2. Limitations. This review only explored four databases (ACM Digital Library, Scopus, SpringerLink, and IEEE). Other databases, such as Google Scholar, Elsevier, and Taylor and Francis, were not included. Exploring a limited number of databases may mean that relevant literature is absent from this review. For example, if the researchers had used Google Scholar, it would show results based on the search query, such as the popularity of the article and how many times it has been cited. Instead, the authors' depended on publication type and journal name to determine relevance.

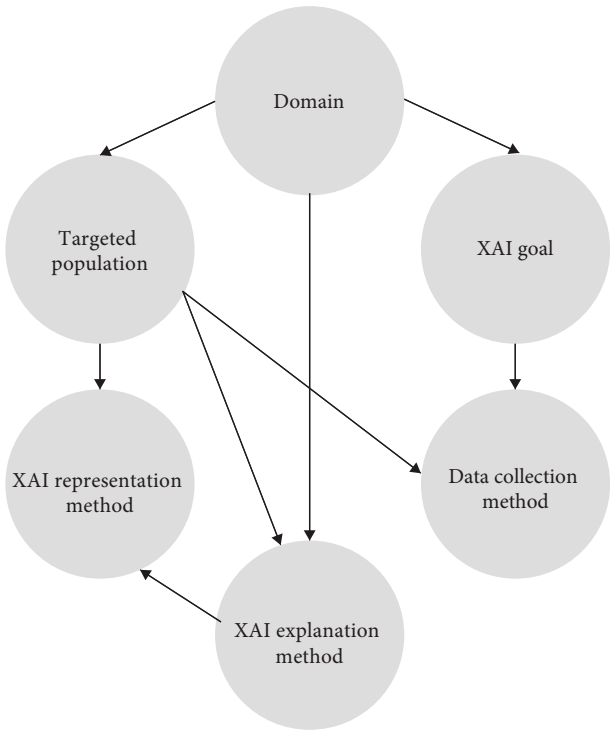

Figure 11: Six pillars to evaluate the achievement of XAI goals following a user-centered approach.

Other databases were not explored due to time constraints, and the search query generated more than 4000 articles, a high number of articles.

Only retrieving studies published in English also means that relevant studies published in other languages may have been missed. The studies included in this review belonged to multiple countries: the United States, Germany, Canada, the United Kingdom, China, India, Italy, the Netherlands, Portugal, Sweden, Singapore, Israel, Korea, Ireland, Australia, Belgium, Finland, France, and Spain. However, there might be relevant studies that explored other populations, making this study's results less comprehensive.

In addition, this review focused on studies where the ML model used was explicitly stated. This approach was followed to explore whether a correlation exists between the ML model and the XAI goals. However, it was concluded that researchers are still researching which ML model works best for many of the XAI goals identified in [84].

8. Lessons Learned

This systematic review has shown that researchers are exploring every XAI goal, who are actively experimenting how it could be achieved across various domains. The pivotal role of the end user is highly important to consider when targeting a specific XAI goal. Evaluating these systems using user-centered evaluation methods is crucial, as it is the end users who will ultimately interact with and benefit from the technology. Consequently, developers and researchers must prioritize the intended audience and ensure the suitability of the XAI system for them, achieving technological advances along with catering users' needs and preferences. Here are three lessons learned from the review to enhance user-centered evaluation of XAI goals.

8.1. Broadening the Scope of the Study. In a research study, precision regarding the area under investigation and the facets being explored is of high importance. However, thoroughly evaluating various aspects within this area may necessitate broadening the scope to evaluate the adequacy of the proposed solution. This has been articulated by Alqaraawi et al. [18], suggesting that a comprehensive assessment often requires considering a broader range of factors to determine the effectiveness of the solution in addressing the identified issues. By widening the scope, researchers can gain a more holistic understanding of the problem and its potential solutions, enabling a more informed and thorough evaluation process.

8.2. The Importance of Carefully Selecting the Targeted Audience. Many aspects of a system designed for use by specific individuals are dependent on the characteristics and preferences of the targeted audience. Consequently, it is crucial to carefully select the intended audience to ensure alignment between the system's design and their capabilities, thereby avoiding challenges during system evaluation. These challenges may include user-related issues concerning XAI explanation methods or XAI representation methods. For instance, a general audience may find excessive explanations overwhelming on the screen, whereas individuals with expertise in a particular area may perceive insufficient information and require more detailed explanations to comprehend the system's predictions.

8.3. User-Centered Validation Measures. It is essential to capitalize on pre-existing validated measures of data collection to investigate domains with a relatively limited number of studies. For example, trustworthiness and informativeness are XAI goals with the highest number of validation measures, primarily consisting of surveys and questionnaires, as explained in Section 4.3.4. The Sankey diagram in Figure 5 also reveals a scarcity of studies in domains such as the educational, social media, and legal domains to evaluate the trustworthiness in the developed XAI systems. Similarly, limited number of studies assessed informativeness of XAI systems in the legal, recruitment, and social media domains. Thus, there exists a significant opportunity for researchers to delve into these domains, given their substantial impact on societal contexts.

As explained above, the facilitation of new research work aimed at evaluating XAI goals is notably expanding based on the existence of prevalidated measures, thus offering a practical area for investigation. Nevertheless, this prevalence of validated tools tends to divert attention from other XAI goals, such as fairness, confidence, and transferability, which lack established measures. This leads to fewer studies focusing on the achievement of these goals, as shown in Section 4.1.5 (Figure 5). So, this review advocates for the allocation of research resources, particularly time, towards the development and validation of assessment tools for these

comparatively underexplored goals. Such endeavors hold promise for enriching our understanding of the efficacy and applicability of XAI systems across diverse contexts and domains.

In conclusion, this review has provided several lessons that, if taken into consideration, have potential to significantly enhance future work. It is important to be precise in terms of the area and aspects to be explored. However, breadth is also required to assess the viability of the XAI solution. For specificity and breadth in research approaches, researchers can attain a comprehensive understanding of XAI systems' efficacy and applicability across diverse contexts and domains, ultimately contributing to advancements in the field. It is also necessary to carefully select the targeted audience to align system design with user capabilities, thus facilitating effective system evaluation. Additionally, utilizing pre-existing validated measures to investigate underexplored domains and validating new measures for less-studied XAI goals is also essential.

9. Conclusion

The studies included in this review were published in a diverse range of 60 publication venues, including conferences such as IUI, CHI, and UMAP; journals like AIJ; and proceedings such as Lecture Notes in Computer Science. The included studies were conducted in 19 countries, with the United States (39.6%), Germany (22.7%), Canada (5%), and Singapore (5%), having the highest frequency of user-centered studies in the field of XAI. The remaining studies were carried out in 17 other countries with lower frequencies. Furthermore, there has been a notable increase in the number of nations conducting XAI studies, with a peak in 2021, with research done in 13 nations, followed by 14 countries in 2022 and Australia and Singapore in 2023. The distribution of publication types among the 101 included studies showed that conference papers were the most common publication venues.

Trustworthiness (42.6%), informativeness (32.6%), and interactivity (10.9%) were the most frequently addressed goals, while other goals like confidence, fairness, accessibility, causality, and transferability had lower frequencies. The research categorized studies based on the XAI goals targeted in different domains. Various domains were covered, including medical, economics, education, social media, legal, recruitment, and entertainment, with some studies conducted in unspecified domains solely for evaluating XAI systems.

In terms of the correlation between XAI goals and the ML models used, there was no significant correlation. But the analysis of ML models used in various AI systems revealed a variety of models and methods employed to achieve XAI goals, with DL-based models being the most commonly used (40.5%), followed by classification models (34.5%) and regression models (14%). Looking at explanation methods, it has been demonstrated that using both local and global explainability techniques in XAI systems improves the explainability of ML and DL models, with a preference for local explanations for inexperienced users. However, combining the two techniques can yield more thorough insights. The limited adoption of global explainability in the examined studies, however, emphasizes the necessity of additional research and development in this field, especially for users with varying degrees of AI proficiency. This underscores the significance of giving equal weight to the creation and application of local and global explanation methods in order to enhance AI systems' explainability and transparency as the field of XAI advances.

In order to comprehensively understand how explainability is studied in user-centered research, this article reviewed user studies conducted in the included papers, examining research methods, study design, participant characteristics and recruitment, data collection methods, evaluation tools, and analysis methods, identifying challenges in user-related, scope-related, and methodology-related aspects. Experimental and user study research methods were commonly used, while mixed methods were less frequent. Crowdsourcing platforms were often employed for participant recruitment. Questionnaires were the primary data collection method, with limited use of interviews and observational methods. Web applications were the predominant form of XAI systems used, and quantitative analysis was commonly used for data analysis. The section is concluded by highlighting the challenges in the various aspects, and the subsequent sections provide detailed exploration of each aspect individually.

To date, no work has been published on the user-centered evaluation of XAI and the interrelated aspects of studies conducted in this field. This systematic literature review focused on three key observations regarding the HCI evaluation techniques used to assess XAI goals applied to evaluate XAI systems in meeting their objectives. Firstly, domain and the XAI goals pursued in a study drive the choice of data collection methods in the included studies. Different methods, such as interviews, surveys, and questionnaires, were utilized to gather comprehensive insights from various user groups. As a common data collection method, surveys and questionnaires emerged as versatile tools for data collection, providing valuable quantitative and qualitative data. Observational methods, such as think-aloud approaches and audio/video recordings, were employed to capture real-time user behavior and offer qualitative insights. However, further research is needed to understand the benefits and limitations of observational methods in different XAI evaluation scenarios, as they have been employed in a few of the selected studies.

Secondly, the recruitment of participants in the analyzed studies predominantly involved easily accessible populations, such as crowd workers, general audiences, and students. While this choice may have practical advantages for researchers, it is important to consider the inclusion of understudied populations to gain a deeper understanding of the perspectives and experiences of the actual target users of XAI systems. Involving a more diverse range of participants can provide valuable insights and enhance the validity and generalizability of the findings.

Thirdly, the choice of XAI representation methods varied based on the data type being analyzed and the expertise

of the participants. Visual representation methods, such as CAMs, were employed in a small percentage of studies, particularly for image-based data analysis. The majority of studies utilized a combination of visual and textual representation methods, incorporating graphs, colors, highlighted text, and charts to enhance textual explanations. These methods are aimed at improving explainability and transparency in user-centered XAI experiences.

The primary conclusion drawn from these observations is the interrelationship among the targeted domain, the intended XAI goal, the targeted population, the data collection methods, and the explanation and representation methods used to evaluate XAI goal achievement. This interdependency highlights the significance of establishing a well-defined framework encompassing these six pillars prior to conducting a user-centered XAI study.

Overall, this review highlights the importance of user-centered data collection approaches for evaluating XAI systems effectively. It offers insights into the selection of data collection methods, the inclusion of understudied populations, and the choice of XAI representation methods. Based on the observations made in this review, it is recommended that researchers consider the specific goals and requirements of their study domains. It is also recommended that they strive for the recruitment of participants who are actually the intended audience for an XAI system and explore a range of data collection and representation methods to enhance the evaluation of XAI systems. By adopting user-centered approaches, researchers can contribute to the development and improvement of XAI systems that meet the needs and expectations of their targeted users.

Data Availability Statement

Data extraction sheet of the included articles and the stages of the systematic literature review is available upon request.

Conflicts of Interest

Authors declare that they have no conflict of interest.

Author Contributions

Noor Al-Ansari was responsible for the design and development of the concept, data acquisition, analysis, interpretation, and drafting the manuscript, as well as writing and revising it. Dena Al-Thani was responsible for the conception and design development, data screening, and revising the manuscript. Reem Al-Mansoori was responsible for data screening.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. The supporting information provides readers with access to appendices, which consists of tables that contribute to a comprehensive understanding of the research presented. It also provides access to the PRISMA checklist. 1. *Appendices* provide detailed tables listing study IDs of the included studies for different purposes such as the following: A. Publication year of the included studies (*Appendix A*). B. Publication venues of the included studies (*Appendix B*). C. Countries of the included studies (*Appendix C*). D. Publication type of the included studies (*Appendix D*). E. XAI goals of the included studies (*Appendix E*). F. Domains observed in the included studies (*Appendix F*). G. Research methods used in the included studies (*Appendix G*). H. Participants of the included studies (*Appendix H*). I. Data collection methods of the included studies (*Appendix I*). J. Analysis methods followed in the included studies (*Appendix J*). 2. *PRISMA checklist* outlines the methodological components and reporting standards adhered to in the systematic review conducted for this study. It ensures transparency and completeness in reporting the review process and findings. *(Supporting Information)*

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